

Topic 1: Two Pointer

Subtopic 1.1: Opposite Ends (Left + Right Pointers)

1. **LC #344 - Reverse String** (Easy)
2. **LC #125 - Valid Palindrome** (Easy)
3. **LC #977 - Squares of a Sorted Array** (Easy)
4. **LC #167 - Two Sum II - Input Array Is Sorted** (Medium)
5. **LC #11 - Container With Most Water** (Medium)
6. **LC #15 - 3Sum** (Medium)
7. **LC #16 - 3Sum Closest** (Medium)
8. **LC #18 - 4Sum** (Medium)
9. **LC #611 - Valid Triangle Number** (Medium)
10. **LC #42 - Trapping Rain Water** (Hard)

Subtopic 1.2: Same Direction (Fast & Slow Pointers)

1. **LC #26 - Remove Duplicates from Sorted Array** (Easy)
2. **LC #27 - Remove Element** (Easy)
3. **LC #283 - Move Zeroes** (Easy)
4. **LC #392 - Is Subsequence** (Easy)
5. **LC #80 - Remove Duplicates from Sorted Array II** (Medium)
6. **LC #88 - Merge Sorted Array** (Easy)
7. **LC #844 - Backspace String Compare** (Easy)
8. **LC #189 - Rotate Array** (Medium)
9. **LC #360 - Sort Transformed Array** (Medium)
10. **LC #287 - Find the Duplicate Number** (Medium)

Subtopic 1.3: Partition / Dutch National Flag

1. **LC #905 - Sort Array By Parity** (Easy)
2. **LC #922 - Sort Array By Parity II** (Easy)
3. **LC #75 - Sort Colors** (Medium)
4. **LC #2161 - Partition Array According to Given Pivot** (Medium)

5. **LC #86 - Partition List** (Medium)
6. **LC #2149 - Rearrange Array Elements by Sign** (Medium)
7. **LC #31 - Next Permutation** (Medium)
8. **LC #148 - Sort List** (Medium)
9. **LC #324 - Wiggle Sort II** (Medium)
10. **LC #969 - Pancake Sorting** (Medium)

Topic 2: Sliding Window

Subtopic 2.1: Fixed Size

1. **LC #643 - Maximum Average Subarray I** (Easy)
2. **LC #1343 - Number of Sub-arrays of Size K and Average Greater than or Equal to Threshold** (Medium)
3. **LC #1052 - Grumpy Bookstore Owner** (Medium)
4. **LC #1423 - Maximum Points You Can Obtain from Cards** (Medium)
5. **LC #438 - Find All Anagrams in a String** (Medium)
6. **LC #567 - Permutation in String** (Medium)
7. **LC #1100 - Find K-Length Substrings With No Repeating Characters** (Medium)
8. **LC #1456 - Maximum Number of Vowels in a Substring of Given Length** (Medium)
9. **LC #2090 - K Radius Subarray Averages** (Medium)
10. **LC #2461 - Maximum Sum of Distinct Subarrays With Length K** (Medium)

Subtopic 2.2: Variable Size (Expand-Shrink)

1. **LC #209 - Minimum Size Subarray Sum** (Medium)
2. **LC #3 - Longest Substring Without Repeating Characters** (Medium)
3. **LC #1004 - Max Consecutive Ones III** (Medium)
4. **LC #424 - Longest Repeating Character Replacement** (Medium)
5. **LC #904 - Fruit Into Baskets** (Medium)
6. **LC #713 - Subarray Product Less Than K** (Medium)
7. **LC #1358 - Number of Substrings Containing All Three Characters** (Medium)
8. **LC #1248 - Count Number of Nice Subarrays** (Medium)

9. **LC #992 - Subarrays with K Different Integers** (Hard)

10. **LC #76 - Minimum Window Substring** (Hard)

Subtopic 2.3: Monotonic Window

1. **LC #239 - Sliding Window Maximum** (Hard)

2. **LC #1438 - Longest Continuous Subarray With Absolute Diff Less Than or Equal to Limit** (Medium)

3. **LC #862 - Shortest Subarray with Sum at Least K** (Hard)

4. **LC #1696 - Jump Game VI** (Medium)

5. **LC #918 - Maximum Sum Circular Subarray** (Medium)

6. **LC #496 - Next Greater Element I** (Easy)

7. **LC #503 - Next Greater Element II** (Medium)

8. **LC #739 - Daily Temperatures** (Medium)

9. **LC #901 - Online Stock Span** (Medium)

10. **LC #84 - Largest Rectangle in Histogram** (Hard)

Topic 3: Prefix Based

Subtopic 3.1: Prefix Sum

1. **LC #1480 - Running Sum of 1d Array** (Easy)

2. **LC #724 - Find Pivot Index** (Easy)

3. **LC #1991 - Find the Middle Index in Array** (Easy)

4. **LC #303 - Range Sum Query - Immutable** (Easy)

5. **LC #560 - Subarray Sum Equals K** (Medium)

6. **LC #525 - Contiguous Array** (Medium)

7. **LC #974 - Subarray Sums Divisible by K** (Medium)

8. **LC #523 - Continuous Subarray Sum** (Medium)

9. **LC #238 - Product of Array Except Self** (Medium)

10. **LC #1094 - Car Pooling** (Medium)

Subtopic 3.2: Prefix XOR

1. **LC #1310 - XOR Queries of a Subarray** (Medium)

2. **LC #1442 - Count Triplets That Can Form Two Arrays of Equal XOR** (Medium)
3. **LC #2429 - Minimize XOR** (Medium)
4. **LC #2433 - Find The Original Array of Prefix XOR** (Medium)
5. **LC #1734 - Decode XORed Permutation** (Medium)
6. **LC #1829 - Maximum XOR for Each Query** (Medium)
7. **LC #2527 - Find Xor-Beauty of Array** (Medium)
8. **LC #268 - Missing Number** (Easy)
9. **LC #136 - Single Number** (Easy)
10. **LC #137 - Single Number II** (Medium)

Subtopic 3.3: 2D Prefix

1. **LC #304 - Range Sum Query 2D - Immutable** (Medium)
2. **LC #1277 - Count Square Submatrices with All Ones** (Medium)
3. **LC #1314 - Matrix Block Sum** (Medium)
4. **LC #1504 - Count Submatrices With All Ones** (Medium)
5. **LC #221 - Maximal Square** (Medium)
6. **LC #1074 - Number of Submatrices That Sum to Target** (Hard)
7. **LC #363 - Max Sum of Rectangle No Larger Than K** (Hard)
8. **LC #85 - Maximal Rectangle** (Hard)
9. **LC #661 - Image Smoother** (Easy)
10. **LC #1292 - Maximum Side Length of a Square with a Sum Less than or Equal to Threshold** (Medium)

Topic 4: Kadane's / Subarray

Subtopic 4.1: Max Subarray Sum (Kadane's Algorithm)

1. **LC #53 - Maximum Subarray** (Medium)
2. **LC #918 - Maximum Sum Circular Subarray** (Medium)
3. **LC #1749 - Maximum Absolute Sum of Any Subarray** (Medium)
4. **LC #1186 - Maximum Subarray Sum with One Deletion** (Medium)
5. **LC #152 - Maximum Product Subarray** (Medium)

6. **LC #2321 - Maximum Score Of Spliced Array** (Hard)
7. **LC #2606 - Find the Substring With Maximum Cost** (Medium)
8. **LC #1191 - K-Concatenation Maximum Sum** (Medium)
9. **LC #121 - Best Time to Buy and Sell Stock** (Easy)
10. **LC #122 - Best Time to Buy and Sell Stock II** (Medium)

Subtopic 4.2: Max Product Subarray

1. **LC #152 - Maximum Product Subarray** (Medium)
2. **LC #1567 - Maximum Length of Subarray With Positive Product** (Medium)
3. **LC #1464 - Maximum Product of Two Elements in an Array** (Easy)
4. **LC #628 - Maximum Product of Three Numbers** (Easy)
5. **LC #713 - Subarray Product Less Than K** (Medium)
6. **LC #1352 - Product of the Last K Numbers** (Medium)
7. **LC #2110 - Number of Smooth Descent Periods of a Stock** (Medium)
8. **LC #2260 - Minimum Consecutive Cards to Pick Up** (Medium)
9. **LC #198 - House Robber** (Medium)
10. **LC #213 - House Robber II** (Medium)

Subtopic 4.3: Subarray with Given XOR / Sum

1. **LC #560 - Subarray Sum Equals K** (Medium)
2. **LC #974 - Subarray Sums Divisible by K** (Medium)
3. **LC #523 - Continuous Subarray Sum** (Medium)
4. **LC #1310 - XOR Queries of a Subarray** (Medium)
5. **LC #1442 - Count Triplets That Can Form Two Arrays of Equal XOR** (Medium)
6. **LC #1658 - Minimum Operations to Reduce X to Zero** (Medium)
7. **LC #2024 - Maximize the Confusion of an Exam** (Medium)
8. **LC #1248 - Count Number of Nice Subarrays** (Medium)
9. **LC #930 - Binary Subarrays With Sum** (Medium)
10. **LC #1004 - Max Consecutive Ones III** (Medium)

Topic 5: Binary Search

Subtopic 5.1: Binary Search on Index

1. **LC #704 - Binary Search** (Easy)
2. **LC #35 - Search Insert Position** (Easy)
3. **LC #34 - Find First and Last Position of Element in Sorted Array** (Medium)
4. **LC #33 - Search in Rotated Sorted Array** (Medium)
5. **LC #81 - Search in Rotated Sorted Array II** (Medium)
6. **LC #153 - Find Minimum in Rotated Sorted Array** (Medium)
7. **LC #162 - Find Peak Element** (Medium)
8. **LC #852 - Peak Index in a Mountain Array** (Medium)
9. **LC #74 - Search a 2D Matrix** (Medium)
10. **LC #240 - Search a 2D Matrix II** (Medium)

Subtopic 5.2: Binary Search on Answer

1. **LC #875 - Koko Eating Bananas** (Medium)
2. **LC #1011 - Capacity To Ship Packages Within D Days** (Medium)
3. **LC #1283 - Find the Smallest Divisor Given a Threshold** (Medium)
4. **LC #1482 - Minimum Number of Days to Make m Bouquets** (Medium)
5. **LC #1552 - Magnetic Force Between Two Balls** (Medium)
6. **LC #410 - Split Array Largest Sum** (Hard)
7. **LC #69 - Sqrt(x)** (Easy)
8. **LC #374 - Guess Number Higher or Lower** (Easy)
9. **LC #278 - First Bad Version** (Easy)
10. **LC #2300 - Successful Pairs of Spells and Potions** (Medium)